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Electric power tool

Abstract

There is provided an electric power tool including: a second voltage supply portion configured to supply, based on a first power supply voltage supplied from a first battery, a second operating voltage for operating a second controller, and configured to supply, based on a second power supply voltage supplied from a second battery, the second operating voltage for operating the second controller; a first voltage line configured to connect a first voltage supply portion and the second voltage supply portion and configured to supply the first power supply voltage; and a backflow suppression circuit configured to permit a current to flow from the first voltage supply portion to the second voltage supply portion in the first voltage line and configured to suppress a current from flowing from the second voltage supply portion to the first voltage supply portion in the first voltage line.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority to Japanese Patent Application No. 2022-064774 filed on Apr. 8, 2022, the contents of which are incorporated herein by way of reference.

TECHNICAL FIELD

(2) The present invention relates to an electric power tool.

BACKGROUND ART

(3) In recent years, there is an electric power tool improved in convenience by mounting a wireless communication device on a main body of the electric power tool.

(4) Patent Literature 1 discloses such an electric power tool on which a wireless communication device can be mounted. The electric power tool includes an electric motor, a communication adapter on which a communication device is mounted, and a battery configured to be detachable.

Patent Literature 1: JP2014-529320A

(5) However, since the electric power tool in Patent Literature 1 includes only a single battery, electric power of the battery is consumed not only by the electric motor but also by the communication device, and a sufficient working time cannot be provided. In addition, when the battery is removed from the main body, communication cannot be performed.

(6) Therefore, as described in Patent Literature 1, addition of a built-in power supply to the electric power tool can also be considered.

(7) However, when a built-in power supply is mounted on the electric power tool, a situation is observed in which even electric power of the built-in power supply sharing a power supply line is consumed for driving the electric motor while the electric motor is driven by being supplied with electric power from the main battery, and the built-in power supply cannot be effectively used.

(8) Therefore, an object of the present invention is to provide an electric power tool in which a plurality of batteries can be mounted and effectively used.

SUMMARY OF INVENTION

(9) There is provided an electric power tool including a power supply device for the electric power tool. The power supply device includes: a first controller configured to control an electric motor mounted in the electric power tool; a first voltage supply portion configured to supply, based on a first power supply voltage supplied from a first battery, a first operating voltage for operating the first controller; a second controller configured to control a communication device mounted in the electric power tool; a second voltage supply portion configured to supply, based on the first power supply voltage supplied from the first battery, a second operating voltage for operating the second controller, and configured to supply, based on a second power supply voltage supplied from a second battery, the second operating voltage for operating the second controller; a first voltage line configured to connect the first voltage supply portion and the second voltage supply portion and configured to supply the first power supply voltage; a second voltage line configured to connect the first voltage supply portion and the second voltage supply portion and configured to supply a second voltage lower than the first power supply voltage; and a backflow suppression circuit configured to permit a current to flow from the first voltage supply portion to the second voltage

supply portion in the first voltage line and configured to suppress a current from flowing from the second voltage supply portion to the first voltage supply portion in the first voltage line.

(10) When the first power supply voltage is applied from the first battery to the first voltage line, the power supply device for the electric power tool may be configured to execute a first operation in which a current flows from the first battery to the second voltage supply portion in the first voltage line and a current flows from the first voltage supply portion to the second voltage supply portion in the second voltage line.

(11) When the first power supply voltage is not applied from the first battery to the first voltage line, the power supply device for the electric power tool may be configured to execute a second operation in which no current flows in the first voltage line and a current flows from the second voltage supply portion to the first voltage supply portion in the second voltage line.

(12) Further, there is provided an electric power tool. The electric power tool includes: the power supply device for the electric power tool as exemplified in the embodiments; the electric motor; the communication device; the second battery; and the first battery configured to be detachably attached to a main body including the power supply device for the electric power tool, the electric motor, the communication device and the second battery.

(13) The “electric power tool” of the present invention refers to a tool that uses electricity as a power source and that is used for work such as machining. The “electric power tool” of the present invention includes, in addition to a product used alone, a component or a module that constitutes a part of another device. For example, the “electric power tool” of the present invention may be a module attached to equipment of a robot arm. Further, the “electric power tool” of the present invention may be a component constituting a part of a machine tool having a plurality of functions.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a perspective view of an electric power tool according to an embodiment.
- (2) FIG. 2 is a cross-sectional view of the electric power tool according to the embodiment.
- (3) FIG. 3 is a block diagram illustrating an electric circuit configuration of the electric power tool according to the embodiment.
- (4) FIG. 4 is a graph illustrating voltages of a first voltage line and a second voltage line in an operating mode of a reinforcing bar binding machine according to the embodiment.

DESCRIPTION OF EMBODIMENTS

- (5) Hereinafter, embodiments of the present invention will be described with reference to the drawings. The following embodiments are examples for explaining the present invention, and it is not intended to limit the present invention only to the embodiments.
- (6) For the sake of convenience, a left-right direction of the paper surface of FIG. 2 may be referred to as a front-rear direction X (an example of a “first direction”), and in particular, a left direction of the paper surface may be referred to as a front side X1 and a right direction of the paper surface may be referred to as a rear side X2; an up-down direction of the paper surface may be referred to as an up-down direction Z (an example of a “second direction”), and in particular, an upper direction of the paper surface may be referred to as an upper side Z1 and a lower direction of the paper surface may be referred to as a lower side Z2; a direction perpendicular to the front-rear direction X and the up-down direction Z may be referred to as a right-left direction Y (an example of a “third direction”), and in particular, a right direction may be referred to as a right side Y1 and a left direction may be referred to as a left side Y2 when facing the front side X1. This is used for the purpose of explaining a relative directional relationship, and does not indicate an absolute direction.
- (7) Hereinafter, an embodiment in which the present invention is applied to a reinforcing bar

binding machine that is an electric power tool will be described. FIG. 1 is a perspective view of a reinforcing bar binding machine **10** that is an electric power tool according to the present embodiment. FIG. 2 is a cross-sectional view obtained by cutting the reinforcing bar binding machine **10**, which is the electric power tool, along a cross section perpendicular to the left-right direction Y.

(8) The present invention can be widely applied to an electric power tool that has a communication function and performs work using an electric motor (motor), and can be applied to, for example, a drill, an impact driver, a nailer, a grinder, a reciprocating saw, and a polisher. The motor may be a brushless motor or a motor with brush.

(9) [Basic Configuration of Electric Power Tool]

(10) The reinforcing bar binding machine **10** according to the present embodiment is configured to bind two reinforcing bars RB or three or more reinforcing bars RB by feeding a wire W outward from an end thereof at the front side X1.

(11) Specifically, the reinforcing bar binding machine **10** includes: a handle **10H** to be gripped by a worker; a magazine **10M** for accommodating the wire W; a wire feeding portion **12** for feeding the wire W outward from the end at the front side X1; a curl forming portion **14** constituting a traveling path of the wire W for winding the wire W around the reinforcing bar RB; a cutting portion **16** for cutting the wire W wound around the reinforcing bar RB; a binding portion **18** for twisting the wire W wound around the reinforcing bar RB; a tool controller that includes a drive controller **22** for controlling both a feeding motor **12M** and a binding motor **18M** that are provided in the wire feeding portion **12** and the binding portion **18**, respectively; and a communication portion **30** that includes a communication device **32** for the reinforcing bar binding machine **10** to communicate with an external device, and a communication controller **34** for controlling the communication device **32**.

(12) In the reinforcing bar binding machine **10** of the present embodiment, the curl forming portion **14**, the cutting portion **16**, the binding portion **18**, the tool controller, and the communication portion **30** constitute a main body **10B** of the reinforcing bar binding machine **10**. The magazine **10M** is provided extending from a lower portion of the main body **10B** at the front side X1 to the lower side Z2. The handle **10H** is provided extending from a lower portion of the main body **10B** at the rear side X2 to the lower side Z2. Thus, the magazine **10M** is provided at the front side X1 with respect to the handle **10H**, and the handle **10H** is provided at the rear side X2 with respect to the magazine **10M**. Further, a lower portion of the magazine **10M** and a lower portion of the handle **10H** are connected to each other. Hereinafter, the configurations will be described.

(13) The reinforcing bar binding machine **10** includes the handle **10H** that extends from the main body **10B** to the lower side Z2. The handle **10H** corresponds to a portion where a worker grips the reinforcing bar binding machine **10**. A lower end of the handle **10H** is formed such that a main battery **10BP** can be detachably attached thereto. A trigger **10HT** is provided on a surface of the handle **10H** facing the front side X1. The reinforcing bar binding machine **10** is configured such that, when the worker presses the trigger **10HT** toward the rear side X2, the tool controller starts a control operation to start a binding operation as will be described later.

(14) The magazine **10M** rotatably and detachably houses a reel RL around which the linear wire W is wound. Here, the reel RL is configured to simultaneously feed one or more wires W. The wire W is a linear body suitable for binding the reinforcing bar RB, such as a long metal wire having flexibility (including a coated metal wire).

(15) The wire feeding portion **12** includes a pair of gears **12G** configured to advance the wire W by rotating in different directions from each other with the wire W sandwiched therebetween, and the feeding motor **12M** (an example of an “electric motor”) configured to drive the gears **12G**. The feeding motor **12M** includes a rotor and a stator. The wire feeding portion **12** is configured to feed the wire W outward by rotating the rotor of the feeding motor **12M** in a forward direction and to pull back the wire W by rotating the rotor in a reverse direction. The tool controller that controls

the feeding motor **12M** of the wire feeding portion **12** will be described later.

(16) The curl forming portion **14** includes a curl guide **14A** configured to curve and curl the wire **W** fed by the wire feeding portion **12**, and an introducing guide **14B** configured to introduce the wire **W** curled by the curl guide **14A** to the binding portion **18**. The curl guide **14A** is configured to curve the wire **W** into a loop shape by advancing the wire **W** along an inner wall surface.

Therefore, when the wire **W** is fed in a state where a plurality of reinforcing bars **RB** are arranged extending in the left-right direction **Y** in a space between the curl guide **14A** and the introducing guide **14B**, it is possible to wind the wire **W** around the reinforcing bars **RB**.

(17) The cutting portion **16** includes a fixed blade, a movable blade that cuts the wire **W** in cooperation with the fixed blade, and a transmission mechanism **16A** that transmits an operation of the binding portion **18** to the movable blade. The cutting portion **16** is configured to cut the wire **W** by a rotational operation of the movable blade with the fixed blade serving as a fulcrum shaft. The transmission mechanism **16A** is configured to transmit the operation of the binding portion **18** to the movable blade and configured to rotate the movable blade in conjunction with a binding operation of the binding portion **18**. Therefore, the transmission mechanism **16A** rotates the movable blade in conjunction with the operation of the binding portion **18**, whereby the wire **W** can be cut at a predetermined timing as will be described later.

(18) The binding portion **18** includes a pair of hooks **18H** configured to be openable and closable in order to clamp the wire **W**, a rotary shaft for rotating the pair of hooks **18H** with the front-rear direction **X** as a rotation axis, a speed reducer configured to move the rotary shaft in a rotation axis direction (front-rear direction **X**) and configured to rotate the rotary shaft moved to the front side **X1**, and the binding motor **18M** configured to be rotatable about a rotation axis **AX**.

(19) The rotary shaft of the binding portion **18** rotates in a forward direction when the binding motor **18M** rotates in a forward direction. A sliding portion is provided around the rotary shaft, and the sliding portion is configured to move toward the front side **X1** when the rotary shaft rotates in the forward direction. The wire **W** fed by the feeding motor **12M** in a state where the pair of hooks **18H** are opened proceeds while being curved along inner wall surfaces of the curl guide **14A** and the introducing guide **14B**, and a leading end of the wire **W** passes through a gap between the pair of hooks **18H**. In this state, when the binding motor **18M** rotates in the forward direction and the rotary shaft rotates in the forward direction, the sliding portion moves to the front side **X1** and the pair of hooks **18H** are closed. Therefore, the pair of hooks **18H** can clamp the wire **W**. When the binding motor **18M** further rotates in the forward direction and the rotary shaft rotates in the forward direction to move the sliding portion to the front side **X1**, the movable blade is rotated by the transmission mechanism **16A** to cut the wire **W**. When the binding motor **18M** further rotates in the forward direction, the binding portion **18** bends the leading end of the cut wire **W** in a state where the wire **W** is sandwiched by the pair of hooks **18H**. When the binding motor **18M** further rotates in the forward direction, the rotary shaft rotates together with the sliding portion with the front-rear direction **X** as a rotation axis. The rotary shaft rotates in the state where the wire **W** is sandwiched by the pair of hooks **18H**, whereby the pair of hooks **18H** twist the wire **W**.

(20) [Electric Circuit Configuration of Electric Power Tool]

(21) FIG. 3 is a block diagram illustrating an electric circuit configuration of the reinforcing bar binding machine **10** according to the present embodiment. In the electric circuit configuration of the reinforcing bar binding machine **10**, a configuration for controlling the feeding motor **12M**, a configuration for controlling the communication device **32**, and a configuration for supplying electric power (voltage) to these configurations constitute a power supply device **40** of the present embodiment.

(22) Specifically, the power supply device **40** according to the present embodiment includes: a battery connection portion **40CN** for receiving supply of electric power (voltage) from the battery **10BP** (which may be referred to as a “driving battery” or a “first battery”); a power supply switch **40S** for turning on or off supply of electric power (voltage) supplied from the battery connection

portion **40CN**; a first electric power controller **41PC** (an example of a “first voltage supply portion”) that receives supply of electric power (voltage) from the battery **10BP** via the power supply switch **40S** and that supplies, based on a first power supply voltage supplied from the battery **10BP**, a first operating voltage for operating the drive controller **22** (an example of a “first controller”); the drive controller **22** that operates based on the first operating voltage and generates a control signal for controlling the feeding motor **12M**; and a motor controller **24** that controls a current flowing through the stator of the feeding motor **12M** based on the control signal generated by the drive controller **22**.

(23) The battery **10BP** is configured to supply electric power for operating at least the drive controller **22**, the motor controller **24**, the feeding motor **12M**, the binding motor **18M**, the communication controller **34**, a wireless communication device **32**, and a position information acquisition portion **36**, the last three of which are to be described later. The battery **10BP** is, for example, a rechargeable lithium ion secondary battery, and has a predetermined rated capacity, rated voltage, and rated current. For example, the battery **10BP** has a rated capacity of 5.0 Ah, and is configured to supply a DC voltage having a rated value of 14.4 V. However, as will be described later, the DC voltage supplied from the battery **10BP** decreases gradually accompanying power consumption of the battery **10BP**.

(24) The battery connection portion **40CN** receives supply of the DC voltage from the battery **10BP** and supplies the DC voltage to the first electric power controller **41PC**. The power supply device **40** includes a first voltage line **41** that connects the battery connection portion **40CN** and the first electric power controller **41PC**, and the first power supply voltage of DC supplied from the battery **10BP** is applied to the first voltage line **41**.

(25) According to an operation of a main power supply switch (not illustrated) of the reinforcing bar binding machine **10**, the power supply switch **40S** turns on or off the supply of electric power (voltage) that is supplied from the battery **10BP** to the first electric power controller **41PC** via the battery connection portion **40CN**. Specifically, when a worker turns off the main power supply switch, the power supply switch **40S** cuts off (turns off) the supply of electric power (voltage) from the battery **10BP** to the first electric power controller **41PC**, and when the worker turns on the main power supply switch, the power supply switch **40S** permits (turns on) the supply of electric power (voltage) from the battery **10BP** to the first electric power controller **41PC**.

(26) The first electric power controller **41PC** generates, based on the first power supply voltage supplied from the battery **10BP**, a voltage for operating circuit elements including the drive controller **22**, and supplies the voltage to the circuit elements and the like. The first electric power controller **41PC** (first voltage supply portion) is, for example, a circuit. For example, the first electric power controller **41PC** is configured to generate, based on the first power supply voltage of 14.4 V supplied from the battery **10BP**, a voltage of 3.3 V (an example of a “first operating voltage”) that is an operating voltage of the drive controller **22**, and configured to supply the voltage of 3.3 V to the drive controller **22**, and configured to further supply the first power supply voltage (14.4 V) to the motor controller **24**, a motor controller **26** and stators of respective motors. Here, supplying a voltage includes generating and supplying a voltage, and passing through and supplying a voltage without generating the voltage. The first electric power controller **41PC** is configured to further generate an intermediate voltage higher than the first operating voltage and lower than the first power supply voltage and configured to supply the intermediate voltage to different circuit elements. The first electric power controller **41PC** may include a booster circuit that is configured to generate a voltage higher than the first power supply voltage and configured to supply the generated voltage to different circuit elements.

(27) The drive controller **22** operates based on a voltage of 3.3 V, for example, to generate a control signal for controlling the feeding motor **12M** and supplies the control signal to the motor controller **24**. In addition, the drive controller **22** is configured to control other actuators and the like of the reinforcing bar binding machine **10**. Further, the drive controller **22** supplies the first power supply

voltage, which is supplied from the first electric power controller **41PC**, to (for example, a positive power supply line of) the motor controller **24**. The drive controller **22** is further configured to receive a signal detecting that the trigger **10HT** is pressed, and configured to start a motor control operation based on reception of the signal. In addition, the drive controller **22** may be configured to receive a signal indicating a temperature of the electric power tool (reinforcing bar binding machine **10**) from a thermistor and configured to control the feeding motor **12M** based on reception of the signal. For example, the drive controller **22** may generate different control signals in cases of a relatively high temperature and a relatively low temperature of the electric power tool and may supply the control signals to the motor controller **24**.

(28) The drive controller **22** may include a single or a plurality of processors implemented by an integrated circuit (IC), and a memory (including a non-volatile semiconductor memory storing information in a non-transitory manner) storing firmware that is executed by the processor and that includes a computer command for executing various types of processing described in the present embodiment. The drive controller **22** (first controller) is, for example, a circuit. The drive controller **22** may be implemented by an IC such as an ASIC, an FPGA, a microcontroller, or the like. The drive controller **22** also functions as a part of the tool controller.

(29) The motor controller **24** controls a current flowing through the stator of the feeding motor **12M** based on a control signal generated by the drive controller **22**. For example, the motor controller **24** may include a plurality of (for example, six) semiconductor elements connected in a three-phase bridge manner between a positive power supply line and a negative power supply line of a ground (reference potential), and a driver circuit for generating and supplying a gate signal (or a base signal) to a gate (or a base) of each semiconductor element.

(30) The feeding motor **12M**, which is an electric motor according to the present embodiment, includes, for example, a stator including three-phase windings connected to three-phase outputs of the motor controller **24**, and a rotor configured to be rotatable in both forward and reverse directions according to a rotating magnetic field generated by a current flowing through the windings of the stator. The feeding motor **12M** may further include, for example, a Hall element for detecting a position of the rotor, and the drive controller **22** may be configured to receive a position signal from the Hall element and configured to generate a control signal based on the position signal.

(31) Similarly, although a detailed description is omitted, the power supply device **40** includes, for the binding motor **18M**, a drive controller that generates a control signal for controlling the binding motor **18M** and supplies the control signal to the motor controller **26** of the binding motor **18M**, and the motor controller **26** that controls a current flowing through the stator of the binding motor **18M** based on the control signal generated by the drive controller. The drive controller of the binding motor **18M** and the drive controller **22** of the feeding motor **12M** may be provided in the same semiconductor chip.

(32) The power supply device **40** further includes a first circuit board **41PCB** on which at least the first electric power controller **41PC**, the drive controller **22**, the motor controller **24**, and the motor controller **26** are mounted. The first circuit board **41PCB** is provided with a first connection portion **41CN** (an example of a “first connector”) for connecting to a second circuit board **42PCB** described later via a cable **40CB**. As illustrated in FIG. 3, the first voltage line **41** includes a first wiring **41A** connecting the battery connection portion **40CN** and the first electric power controller **41PC**, and a second wiring **41B** branching from the first wiring **41A** and connected to the first connection portion **41CN**. The power supply device **40** further includes a second voltage line **42** to which the first operating voltage (3.3 V) generated by the first electric power controller **41PC** can be applied and that connects the first electric power controller **41PC** and the first connection portion **41CN**. The first circuit board **41PCB** in FIG. 3 is drawn conceptually, and the actual first circuit board **41PCB** is formed in a rectangular shape having two parallel long sides and two parallel short sides connecting ends of the long sides.

(33) With the above configuration, it is possible to supply electric power for implementing the functions of the electric power tool. Next, a configuration related to a communication function of the electric power tool will be described.

(34) As illustrated in FIG. 3, the power supply device **40** includes: a second battery **42BP** for communication (also may be referred to as a “communication battery”); a second electric power controller **42PC** (an example of a “second voltage supply portion”) that is configured to supply, based on the first power supply voltage supplied from the main battery **10BP**, a second operating voltage for operating the communication controller **34** (an example of a “second controller”), and that is configured to supply, based on a second power supply voltage supplied from the second battery **42BP** in cases such as one where the main battery **10BP** is removed, the second operating voltage for operating the communication controller **34**; and the communication controller **34** that controls the wireless communication device **32** (an example of the “communication device **32**”) and the position information acquisition portion **36** based on the second operating voltage supplied from the second electric power controller **42PC**.

(35) The communication portion **30** of the electric power tool includes the position information acquisition portion **36** that acquires position information of the electric power tool and supplies the position information to the communication controller **34**, and the wireless communication device **32** for wirelessly transmitting information to and receiving information from an external device.

(36) The position information acquisition portion **36** includes, for example, an antenna configured to receive a signal from a positioning satellite of GPS (or another GNSS such as GLONASS), and a reception circuit that acquires position information of the electric power tool based on the signal received by the antenna.

(37) For example, the wireless communication device **32** includes: an antenna that is configured to transmit information to and receive information from a remote base station according to a predetermined standard by using a frequency band of a licensed band or an unlicensed band based on the LPWA technology; an RFIC that is configured to demodulate an analog signal received by the antenna and configured to supply the demodulated analog signal to a baseband IC, and that is configured to modulate a signal supplied from the baseband IC into an analog signal and configured to transmit the analog signal from the antenna; and the baseband IC that decodes or encodes a signal acquired from the RFIC according to a protocol defined by the standard and transmits and receives information. The communication device **32** may be compatible with a short-distance wireless communication system such as Bluetooth (registered trademark) or a wireless LAN.

(38) With the above-described configuration, for example, the communication controller **34** is configured to provide the position information of the electric power tool to an external device by transmitting, via the wireless communication device **32**, the position information acquired by the position information acquisition portion **36**. The communication controller **34** (second controller) is, for example, a circuit. In the present embodiment, the communication controller **34**, the RFIC, and the baseband IC may be stacked in the same semiconductor package. The communication controller **34** and the baseband IC may be formed of the same semiconductor chip.

(39) The power supply device **40** further includes the second circuit board **42PCB** on which at least the second electric power controller **42PC**, the communication controller **34**, the communication device **32**, and the position information acquisition portion **36** are mounted. The second circuit board **42PCB** is provided with a second connection portion **42CN** (an example of a “second connector”) for connecting to the first circuit board **41PCB** via the cable **40CB**. As illustrated in FIG. 3, with respect to the first voltage line **41**, the first wiring **41A** and the second wiring **41B** are formed on the first circuit board **41PCB**, and a third wiring **41C** electrically connected to the first wiring **41A** and the second wiring **41B** via the connectors is formed on the second circuit board **42PCB**. With respect to the second voltage line **42**, a wiring connecting the first connection portion **41CN** and the first electric power controller **41PC** is formed on the first circuit board **41PCB**, and a

wiring electrically connected to the former wiring via the connectors and connected to the second electric power controller **42PC** is formed on the second circuit board **42PCB**.

(40) The second battery **42BP** further provided in the power supply device **40** supplies electric power for operating at least the communication controller **34**, the wireless communication device **32**, and the position information acquisition portion **36**. The second battery **42BP** is, for example, a rechargeable lithium ion secondary battery, and has a predetermined rated capacity, rated voltage, and rated current. For example, the second battery **42BP** has a smaller rated capacity than the main battery **10BP**, and is configured to supply a DC voltage having a rated value of 3.6 V (an example of the “second power supply voltage”). In addition, since the second battery **42BP** is housed in a housing constituting the main body **10B** of the electric power tool, unlike the main battery **10BP**, the second battery **42BP** is not provided to be easily detachable but is integrally fixed to the second circuit board **42PCB**. The second battery **42BP** may not be integrally fixed to the second circuit board **42PCB**, and may be detachably provided on the second circuit board **42PCB**, for example.

(41) The second electric power controller **42PC** is configured to generate, based on a power supply voltage supplied from the battery **10BP**, a voltage for operating circuit elements including the communication controller **34** and the drive controller **22**, and configured to supply the voltage to the circuit elements and the like. The second electric power controller **42PC** (second voltage supply portion) is, for example, a circuit. For example, the second electric power controller **42PC** generates, based on a power supply voltage of 14.4 V supplied from the battery **10BP**, a voltage (an example of a “second operating voltage”) of 3.3 V that is an operating voltage of the communication controller **34**, and supplies the voltage to the communication controller **34** by a third voltage line **43** connecting the second electric power controller **42PC** and the communication controller **34**, and similarly, generates a predetermined operating voltage and supplies the predetermined operating voltage to the position information acquisition portion **36** and the wireless communication device **32**.

(42) In addition, the second electric power controller **42PC** is configured to generate, based on a DC voltage of 3.6 V corresponding to the second power supply voltage supplied from the second battery **42BP**, an operating voltage for operating circuit elements and the like, and configured to supply the generated operating voltage to the circuit elements including the communication controller **34** and the drive controller **22**. Here, the second electric power controller **42PC** includes a booster circuit that is configured to generate a voltage higher than the second power supply voltage in order to operate the antenna. However, the reinforcing bar binding machine **10** may not necessarily include the booster circuit. In the present embodiment, since an operating voltage of the antenna is higher than the second power supply voltage, the reinforcing bar binding machine **10** includes a booster circuit such as a charge pump circuit. However, for example, when a circuit element to be driven by electric power of the second battery **42BP** is selected or the second power supply voltage is set such that an operating voltage of the circuit element is equal to or lower than the second power supply voltage, the reinforcing bar binding machine **10** may not necessarily include the booster circuit.

(43) With the above configuration, when the main battery **10BP** is removed, the second electric power controller **42PC** operates the drive controller **22**, the communication controller **34**, the position information acquisition portion **36**, and the wireless communication device **32** based on the second power supply voltage supplied from the second battery **42BP**, thereby enabling wireless transmission and reception of information with an external device. Accordingly, even when the main battery **10BP** is removed, the communication controller **34** can provide the position information of the electric power tool to the outside by transmitting, via the wireless communication device **32**, the position information acquired by the position information acquisition portion **36**, and can store update data for updating the firmware of the drive controller **22** that is received via the wireless communication device **32** in, for example, a non-volatile semiconductor memory constituting the drive controller **22**.

(44) Further, the second electric power controller **42PC** is configured to generate, based on the power supply voltage supplied from the main battery **10BP**, a charging voltage for charging the second battery **42BP**, and configured to charge the second battery **42BP**. Accordingly, when the battery **10BP** is attached, the power supply device **40** according to the present embodiment is configured to operate the drive controller **22**, the motor controller **24**, the motor controller **26**, the motors (the feeding motor **12M** and the binding motor **18M**), the communication controller **34**, the position information acquisition portion **36**, and the wireless communication device **32** based on the electric power supplied from the battery **10BP** and configured to charge the second battery **42BP** based on the electric power supplied from the battery **10BP**. When the main battery **10BP** is removed, the power supply device **40** is configured to operate the drive controller **22**, the communication controller **34**, the position information acquisition portion **36**, and the wireless communication device **32** based on the electric power supplied from the second battery **42BP**. As described above, since supplying a voltage includes passing through and supplying a voltage without generating the voltage, a voltage serving as a power supply may be supplied to the communication controller **34** and the like by directly connecting a wiring to which an output voltage from the second battery **42BP** is applied, to a power supply terminal of the communication controller **34** and the like, or the voltage serving as a power supply may be supplied to the drive controller **22** and the like by directly connecting a wiring to which an output voltage from the second electric power controller **42PC** is applied, to a power supply terminal of the drive controller **22** and the like.

(45) The power supply device **40** further includes the second circuit board **42PCB** on which at least the second electric power controller **42PC**, the communication controller **34**, the position information acquisition portion **36**, and the wireless communication device **32** are mounted. The second circuit board **42PCB** is provided with a second connection portion **42CN** (an example of a “second connector”) for connecting to the first circuit board **41PCB** via the cable **40CB**.

(46) As illustrated in FIG. 3, the first voltage line **41** to which the power supply voltage from the battery **10BP** is applied not only supplies a voltage to the first electric power controller **41PC** via the power supply switch **40S**, but also supplies a voltage to the second power controller **42PC** by including a wiring that is connected to the second electric power controller **42PC** via the first connection portion **41CN** of the first circuit board **41PCB**, the cable **40CB**, and the second connection portion **42CN** of the second circuit board **42PCB**.

(47) Further, the second voltage line **42** to which the first operating voltage (3.3 V) generated by the first electric power controller **41PC** is applied, includes a wiring connected to the second electric power controller **42PC** and the communication controller **34** via the connection portion of the first circuit board **41PCB**, the cable **40CB**, and the connection portion of the second circuit board **42PCB**. Accordingly, when the main battery **10BP** is attached, the second voltage line **42** is configured to supply electric power (voltage and current) in a direction from the first electric power controller **41PC** to the second electric power controller **42PC**, and when the main battery **10BP** is removed, the second voltage line **42** is configured to supply electric power (voltage and current) in a direction from the second electric controller **42PC** to the first electric power controller **41PC**.

(48) Here, the third wiring **41C** of the first voltage line **41** provided on the second circuit board **42PCB** is provided with a diode (an example of a “backflow suppression circuit”) having an anode connected to a first electric power controller **41PC** side and a cathode connected to a second electric power controller **42PC** side. The diode serves as a circuit capable of permitting a current to flow from the first electric power controller **41PC** to the second electric power controller **42PC** and suppressing a current from flowing from the second electric power controller **42PC** to the first electric power controller **41PC**. The backflow suppression circuit may be provided in the second wiring **41B** of the first voltage line **41** provided on the first circuit board **41PCB**.

(49) The inventors of the present application focused on a fact that in a case where a plurality of batteries **10BP** are mounted on the electric power tool, electric power of the second battery **42BP** is

also unintentionally consumed when the electric motors and the communication device **32** are driven by being supplied with electric power from the main battery **10BP**, and conceived of providing a backflow suppression circuit, and verified that power consumption of the second battery **42BP** can be suppressed by the backflow suppression circuit.

(50) That is, in a case of a configuration in which no backflow suppression circuit is provided, a DC voltage supplied from the main battery **10BP** and applied to the first voltage line **41** decreases gradually accompanying the use of the electric power tool, and for example, a DC voltage lower than the original 14.4 V is supplied. When the use of the electric motor (for example, the feeding motor **12M**) of the electric power tool is still continued even though the DC voltage applied to the first voltage line **41** is decreased, a phenomenon is observed in which electric power is supplied from the second battery **42BP** to the electric motor to compensate for the insufficient electric power. At this time, in the first voltage line **41**, in addition to the current flowing from the main battery **10BP** to the first electric power controller **41PC**, a current flowing from the second battery **42BP** to the first electric power controller **41PC** via the second electric power controller **42PC** is observed. As a result, there is a possibility that the second battery **42BP** cannot be charged as expected, or there is a possibility that sufficient electric power cannot be provided to the communication controller **34** or the like due to a shortage of electric power of the second battery **42BP** and the communication function is impaired when the main battery **10BP** is removed.

(51) Therefore, the inventors of the present application conceived of providing a backflow suppression circuit configured to permit a current to flow from the first electric power controller **41PC** (first voltage supply portion) to the second electric power controller **42PC** (second voltage supply portion) in the first voltage line **41** and configured to suppress a current from flowing from the second electric power controller **42PC** (second voltage supply portion) to the first electric power controller **41PC** (first voltage supply portion).

(52) By providing the backflow suppression circuit, it is possible to suppress the flow from the second battery **42BP** to the first electric power controller **41PC** via the second electric power controller **42PC**, and thus it is possible to effectively use the main battery **10BP** and the second battery **42BP**.

(53) The second voltage line **42** is preferably configured to permit a current to flow from the first electric power controller **41PC** (first voltage supply portion) to the second electric power controller **42PC** (second voltage supply portion) and configured to permit a current to flow from the second electric power controller **42PC** (second voltage supply portion) to the first electric power controller **41PC** (first voltage supply portion).

(54) With such a configuration, the electric power (voltage) can be supplied to the power supply terminal of the drive controller **22** of the first circuit board **41PCB** via the second voltage line **42** that connects the second circuit board **42PCB** which is a communication board, the second connector, the first connector, and the first electric power controller **41PC**.

(55) Therefore, when the main battery **10BP** is removed, the drive controller **22** can be operated. For example, the drive controller **22** can update firmware that is a control program. Accordingly, it is possible to reduce a situation where the work of a worker has to be interrupted for updating the firmware of the drive controller **22**, thereby improving work efficiency of the worker.

(56) The second voltage line **42** may be provided so as to connect the first connector and the power supply terminal of the drive controller **22** of the first circuit board **41PCB** directly without using the first electric power controller **41PC** as an intermediary.

(57) In addition, the power supply device **40** according to the present embodiment has a configuration in which the first circuit board **41PCB** and the second circuit board **42PCB** are connected to each other via the cable **40CB** (including a wiring constituting a part of the first voltage line **41** and a wiring constituting a part of the second voltage line **42**). Therefore, the first circuit board **41PCB** and the second circuit board **42PCB** can be disposed at different positions. For example, the first circuit board **41PCB** may be disposed at the upper side **Z1** of the binding motor

18M such that the circuit board is substantially perpendicular to the up-down direction Z, and the second circuit board **42PCB** may be disposed in the left-right direction Y (for example, at the right side **Y1**) of the binding motor **18M** such that the circuit board is substantially perpendicular to the left-right direction Y. With such a configuration, the two circuit boards can be disposed so as to surround a motor (one of the binding motor **18M** and the feeding motor **12M**) in a state of being substantially perpendicular to each other. Here, the two circuit boards may be arranged such that a distance between a rotation axis of the motor (for example, the rotation axis **AX** of the binding motor **18M**) and the second circuit board **42PCB** is larger than a distance between the rotation axis of the motor and the first circuit board **41PCB**. With such a configuration, the second circuit board **42PCB** can be disposed at a position away from the motor as compared with the first circuit board **41PCB**, and thus it is possible to lower a possibility that noise caused by rotation of the motor has a bad influence on communication; the first circuit board **41PCB** can be disposed at a position closer to the motor as compared with the second circuit board **42PCB**, and thus it is possible to reduce a distance between the motor controller **24** (or the motor controller **26**) and the stator of the motor.

(58) [Operations of Reinforcing Bar Binding Machine]

(59) Hereinafter, operations of the reinforcing bar binding machine **10** will be described. The reinforcing bar binding machine **10** according to the present embodiment is configured to execute operating modes described below. FIG. **4** is a graph illustrating voltages of the first voltage line **41** and the second voltage line **42** in the operating modes of the reinforcing bar binding machine **10** according to the present embodiment.

(60) FIG. **4** also illustrates a supply state of electric power from the second battery **42BP** to the circuit elements on the first circuit board **41PCB**. When electric power is supplied from the second battery **42BP** to the circuit elements on the first circuit board **41PCB**, the electric power in the graph in FIG. **4** is negative. At this time, in the wiring of the second voltage line **42** connecting the first circuit board **41PCB** and the second circuit board **42PCB**, a current flows from the second circuit board **42PCB** to the first circuit board **41PCB**, and the second battery **42BP** supplies electric power to the circuit elements on the first circuit board **41PCB**.

(61) When electric power is supplied from the first battery to the circuit elements on the second circuit board **42PCB**, the electric power in the graph in FIG. **4** is positive. At this time, a current flows from the first circuit board **41PCB** to the second circuit board **42PCB** in at least one of the wiring of the first voltage line **41** connecting the first circuit board **41PCB** and the second circuit board **42PCB** and the wiring of the second voltage line **42** connecting the first circuit board **41PCB** and the second circuit board **42PCB**, and the first battery supplies electric power to the circuit elements on the second circuit board **42PCB**.

(62) When no electric power is supplied from the first battery to the circuit elements on the second circuit board **42PCB** and no electric power is supplied from the second battery **42BP** to the circuit elements on the first circuit board **41PCB**, the electric power in the graph in FIG. **4** is zero. At this time, no current flows through any of the wiring of the first voltage line **41** connecting the first circuit board **41PCB** and the second circuit board **42PCB** and the wiring of the second voltage line **42** connecting the first circuit board **41PCB** and the second circuit board **42PCB**.

(63) However, the reinforcing bar binding machine **10** can execute the operating modes in different orders. The electric power tool of the present invention may be configured to execute any one of the operating modes described below, or may be configured to execute any two or more of the operating modes described below.

(64) [Non-Operating State of Electric Motor and Drive Controller, and Operating State of Communication Device]

(65) When the main battery **10 BP** is removed or when the power supply switch **40S** is turned off, the electric motors are not operated and are in a non-operating state. At this time, the communication device **32** of the electric power tool is operable.

(66) In the operation mode from a time point **t0** to a time point **t1** in FIG. **4**, the electric motors and

the drive controller **22** of the electric power tool are in the non-operating state and the communication device **32** is in the operating state. At this time, since no DC voltage from the first battery is applied to the first voltage line **41**, the first voltage line **41** has a voltage of 0 V (ground level) and no current is flowing in the first voltage line **41**. Further, since it is not necessary to operate the circuit elements on the first circuit board **41PCB** including the drive controller **22**, the second voltage line **42** has a voltage of 0 V (ground level) and no current is flowing in the second voltage line **42**. Since no electric power is supplied from the first battery to the circuit elements on the second circuit board **42PCB** and from the second battery **42BP** to the circuit elements on the first circuit board **41PCB**, the electric power in the graph is zero.

(67) At this time, the second battery **42BP** supplies the second power supply voltage of 3.6 V to the second electric power controller **42PC**, and the second electric power controller **42PC** generates, based on a DC voltage of 3.6 V corresponding to the second power supply voltage supplied from the second battery **42BP**, an operating voltage for operating circuit elements of the communication portion **30** and the like, and applies the operating voltage (3.3 V) of the communication controller **34** to the third voltage line **43**. Similarly, the second electric power controller **42PC** generates an operating voltage for operating the circuit elements such as the antenna of the communication portion **30**, and supplies the operating voltage to the circuit elements including the communication controller **34**. Therefore, the communication portion **30** can be operated.

(68) With the above configuration, even when the main battery **10BP** is removed, the electric power tool can provide the position information of the electric power tool to the outside by transmitting, via the wireless communication device **32**, the position information acquired by the position information acquisition portion **36**. Thus, even when the electric power tool is stolen, the position information of the electric power tool can be acquired.

(69) [Non-Operating State of Electric Motor, and Operating State of Drive Controller and Communication Device]

(70) In the operation mode (an example of a “second operation”) from the time point **t1** to a time point **t2** in FIG. 4, the electric motors of the electric power tool are in the non-operating state, and the drive controller **22** and the communication device **32** are in the operating state. At this time, since no DC voltage from the first battery is applied to the first voltage line **41**, the first voltage line **41** has a voltage of 0 V (ground level) and no current is flowing in the first voltage line **41**. On the other hand, the second electric power controller **42PC** applies the operating voltage (3.3 V) of the drive controller **22** to the second voltage line **42** based on the second power supply voltage supplied from the second battery **42BP**. Similarly, the second electric power controller **42PC** applies the operating voltage (3.3 V) of the communication controller **34** to the third voltage line **43** based on the second power supply voltage supplied from the second battery **42BP**. Similarly, the second electric power controller **42PC** generates an operating voltage for operating the circuit elements such as the antenna of the communication portion **30**, and supplies the operating voltage to the circuit elements including the communication controller **34**. At this time, since electric power is supplied from the second battery **42BP** to the drive controller **22** which is a circuit element on the first circuit board **41PCB**, the electric power in the graph is negative, and a current in a direction from the second circuit board **42PCB** to the first circuit board **41PCB** flows through the wiring of the second voltage line **42** connecting the first circuit board **41PCB** and the second circuit board **42PCB**. Therefore, the drive controller **22** and the communication portion **30** can be operated.

(71) With the above configuration, even when the main battery **10BP** is removed, the electric power tool can receive the update data of the firmware, which is the control program of the drive controller **22**, by using the communication portion **30**, can store the update data in the storage section by using the drive controller **22**, and can update the firmware based on the update data.

(72) [Operating State of Electric Motor, and Operating State of Drive Controller and Communication Device]

(73) In the operating mode (an example of a “first operation”) from the time point **t2** in FIG. 4, the

electric motors, the drive controller **22** and the communication device **32** of the electric power tool are in the operating state. When the worker turns on the main power supply switch to perform work, this operating mode starts. At this time, since a DC voltage from the first battery is applied to the first voltage line **41**, the first voltage line **41** has a voltage of 14.4 V. When the worker presses down the trigger **10HT** in this state, since the motors (the binding motor **18M** and the feeding motor **12M**) start to be driven, a current flows through the first wiring **41A** of the first voltage line **41** toward the stators of the motors via the first electric power controller **41PC**.

(74) Further, the first battery is configured to supply electric power to the communication portion **30**. Specifically, the first electric power controller **41PC** is configured to generate, based on a DC voltage of 14.4 V corresponding to the first power supply voltage supplied from the first battery, a voltage of 3.3 V that is an operating voltage of the communication controller **34**, and configured to supply the generated voltage to the communication controller **34** via the second voltage line **42**.

(75) In addition, when a battery charge level of the second battery **42BP** is low, the first battery is configured to charge the second battery **42BP**. Specifically, the second electric power controller **42PC** is configured to generate, based on a DC voltage of 14.4 V corresponding to the first power supply voltage supplied from the first battery, a charging voltage for charging the second battery **42BP**, and configured to charge the second battery **42BP**.

(76) At this time, since electric power is supplied from the first battery to the communication portion **30** and the second battery **42BP** on the second circuit board **42PCB**, the electric power in the graph is positive, and a current in a direction from the first circuit board **41PCB** to the second circuit board **42PCB** flows through wirings of the first voltage line **41** and the second voltage line **42** connecting the first circuit board **41PCB** and the second circuit board **42PCB**.

(77) Here, since the third wiring **41C** of the first voltage line **41** is provided with a diode (an example of the “backflow suppression circuit”), when the electric power is supplied from the first battery **10BP** to the electric motors to drive the electric motors, it is possible to reduce a situation in which even the electric power of the second battery **42BP** is unintentionally consumed for driving the electric motors.

(78) As described above, according to the electric power tool of the present embodiment, in a case where a plurality of batteries such as a main battery and a sub battery are mounted, a situation can be reduced in which even electric power of the sub battery is consumed for driving an electric motor while the electric motor is driven by being supplied with electric power from the main battery, and each battery can be effectively used.

(79) Various modifications can be made without departing from the gist of the present invention. For example, it is possible to add other known techniques to a part of the constituent elements of an embodiment within the scope of the ordinary creative ability of those skilled in the art. A part of constituent elements of an embodiment may be replaced with other known techniques. For example, the drive controller and the communication controller can operate at different voltages.

Claims

1. An electric power tool comprising: an electric motor; a first controller configured to control the electric motor; a first voltage supply portion configured to supply, based on a first power supply voltage supplied from a first battery, a first operating voltage for operating the first controller; a communication device; a second controller configured to control the communication device; the first battery; a second battery; a second voltage supply portion configured to supply, based on the first power supply voltage supplied from the first battery, a second operating voltage for operating the second controller, and configured to supply, based on a second power supply voltage supplied from the second battery, the second operating voltage for operating the second controller; a first voltage line configured to connect the first voltage supply portion and the second voltage supply portion and configured to supply the first power supply voltage; a second voltage line configured to

- connect the first voltage supply portion and the second voltage supply portion and configured to supply a second voltage lower than the first power supply voltage; and a backflow suppression circuit configured to permit a current to flow from the first voltage supply portion to the second voltage supply portion in the first voltage line and configured to suppress a current from flowing from the second voltage supply portion to the first voltage supply portion in the first voltage line.
2. The electric power tool according to claim 1, wherein the second voltage line is configured to permit a current to flow from the first voltage supply portion to the second voltage supply portion and configured to permit a current to flow from the second voltage supply portion to the first voltage supply portion.
 3. The electric power tool according to claim 1, wherein the first operating voltage, the second operating voltage and the second voltage are the same.
 4. The electric power tool according to claim 1, further comprising: a first circuit board on which the first controller and the first voltage supply portion are mounted; a second circuit board on which the second controller and the second voltage supply portion are mounted; a first connector provided on the first circuit board; a second connector provided on the second circuit board; and a cable configured to connect the first connector and the second connector and including a power supply line constituting at least a part of the first voltage line and a power supply line constituting at least a part of the second voltage line.
 5. The electric power tool according to claim 4, wherein the backflow suppression circuit is provided a wiring of the first voltage line provided on the second circuit board.
 6. The electric power tool according to claim 4, wherein the backflow suppression circuit is provided a wiring of the first voltage line provided on the first circuit board.
 7. The electric power tool according to claim 1, wherein when the first power supply voltage is applied from the first battery to the first voltage line, a first operation is to be executed in which a current flows from the first battery to the second voltage supply portion in the first voltage line and a current flows from the first voltage supply portion to the second voltage supply portion in the second voltage line, and when the first power supply voltage is not applied from the first battery to the first voltage line, a second operation is to be executed in which no current flows in the first voltage line and a current flows from the second voltage supply portion to the first voltage supply portion in the second voltage line.
 8. The electric power tool according to claim 1, wherein the backflow suppression circuit is a diode having an anode connected to side of the first voltage supply portion and a cathode connected to a side of the second voltage supply portion.
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